

AMEESH KARTHIK

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with a focus on embedded systems, hardware-software integration, and automated verification. Passionate about applying analytical problem-solving skills to Design Verification workflows, simulation, and high-performance system architecture.

TECHNICAL SKILLS

Programming Languages: C, C++, Embedded C, Python, Java

Frameworks: Node.js, React, Socket.IO

Databases: MySQL

Libraries: Python ML Libraries, YOLO

Developer Tools: Git, GitHub, VS Code, MATLAB, Proteus, Arduino, ESP32

Cloud: MQTT, Cloud Monitoring

PROJECTS

Plant Disease Detection via Autonomous Bot • **Technologies:** *YOLO, Embedded C, Sensor Integration*

- Engineered an autonomous robotics platform utilizing bio-impedance sensing for real-time plant health diagnostics.
- Integrated YOLO-based AI models to achieve rapid image-based classification with optimized edge processing.
- Reduced latency in disease identification by 20% through efficient hardware-software synchronization.

BrainScore AI – Real-time Multiplayer Quiz Platform • **Technologies:** *React, Node.js, Socket.IO, AI*

- Developed a scalable, full-stack multiplayer quiz application featuring real-time bi-directional data flow using Socket.IO.
- Automated question generation using AI, significantly increasing content variety and decreasing manual input time.
- Deployed a highly responsive web interface on Render and Netlify with optimized load balancing for concurrent users.

EDUCATION

Bachelor of Engineering in Electronics and Communication Engineering - Sahyadri College of Engineering and Management, Mangalore 2023-2027

GPA: 9.0/10. Focused on Embedded Systems, VLSI, and Digital Signal Processing.

CERTIFICATIONS

- Introduction to Python Programming
- Fundamentals of Embedded Systems
- Basics of Machine Learning
- Internet of Things (IoT) Workshop

ACHIEVEMENTS

- Consistent Academic Performer with a 9.0 GPA track record.
- Active contributor to open-source hardware projects and AI-driven smart device research.

LANGUAGES

English, Kannada